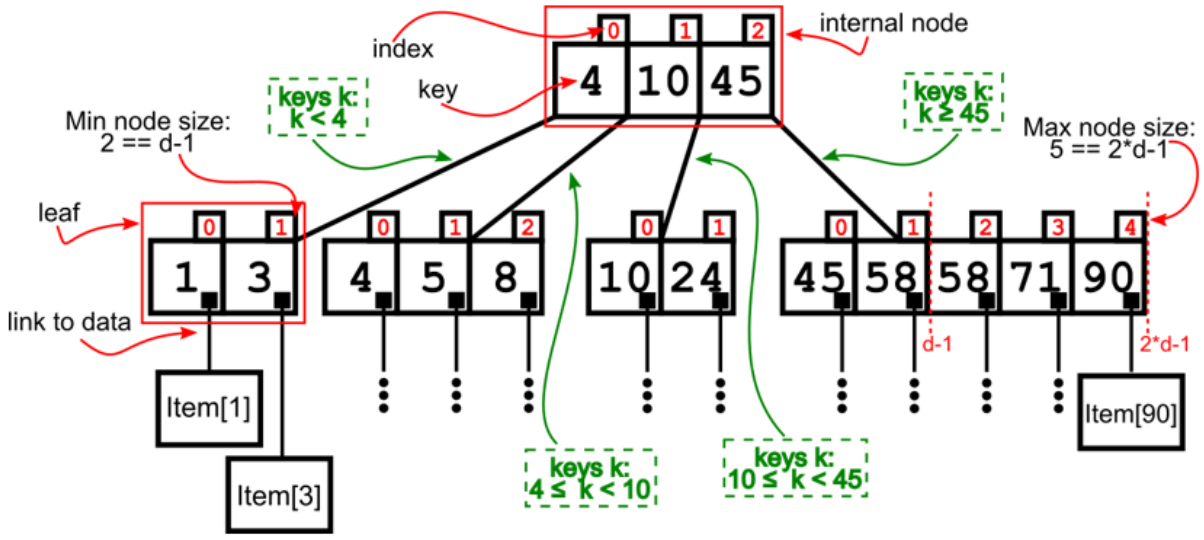


Figure 10.2 B+ tree explained. The example shows a B+ tree with branching factor `d == 3`.

In a B-tree, keys and items are stored in the nodes, each key/item is stored exactly once in a single node, and the whole tree stores exactly `n` keys, if the dataset has `n` items; in B+ tree, instead, internal nodes only contains keys, while only leaves contains pairs each with keys and links to the items; this means that a B+ tree storing `n` items has `n` leaves, and that keys in internal nodes are also stored in all its descendants (see how, in the example in figure 10.2, the keys `4`, `10` and `45` are also stored in leaves).

Storing link to items in the leaves, instead of having the actual items hosted in the tree, serves a double purpose:

1. Nodes are more lightweight and easier to allocate/garbage collect;
2. Allows storing all items in an array, or anyway in a contiguous block of memory, exploiting memory locality of neighboring items.